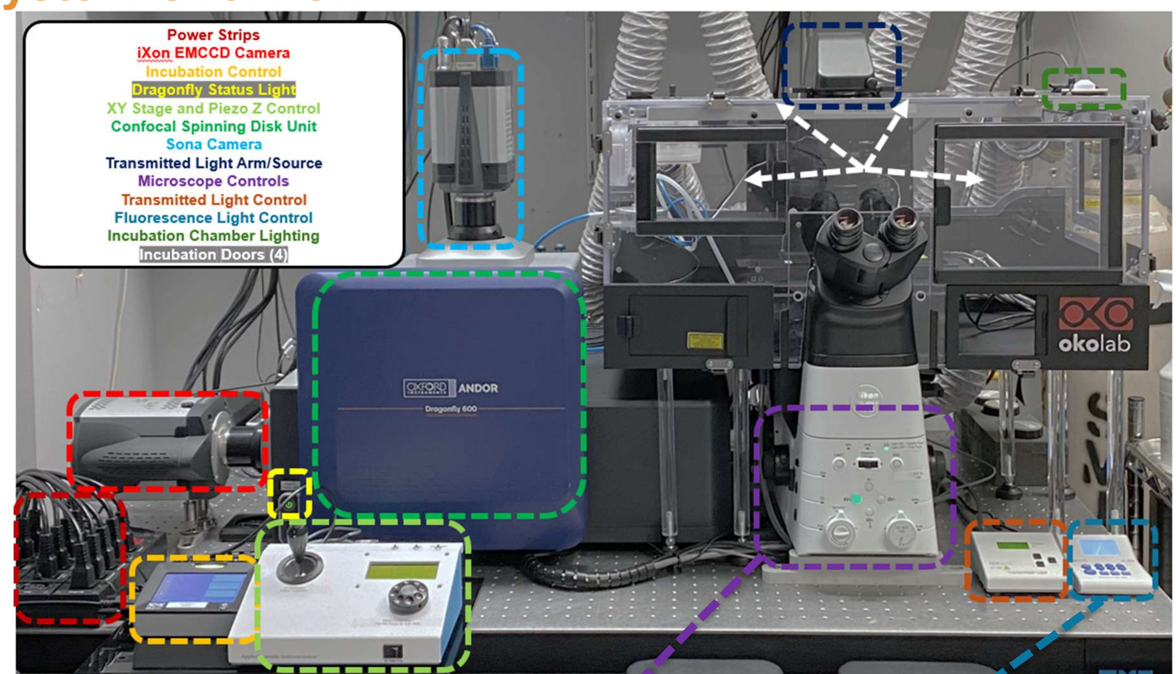
Microscopy CoRE – Andor Dragonfly 620, Basic Operation

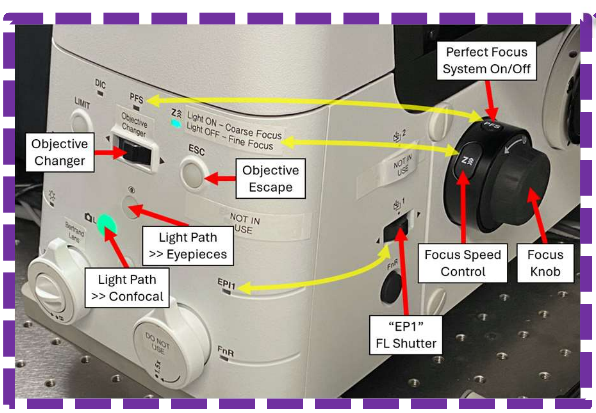
microscopy.core@mssm.edu; 212-241-0400
February 2024

The Andor Dragonfly 620 is a spinning disk confocal with many advanced modalities. It is capable of such techniques as widefield imaging, high-sensitivity or high-contrast spinning disk imaging, TIRF (critical angle, Hi-Lo), STORM (single molecule imaging), photostimulation of specific cellular regions and dual-camera recording. The microscope is enclosed in a live-cell incubator with temperature, CO_2 and O_2 control. The stage features a piezo-Z focus mechanism for high-speed refocusing across multiple specimen fields as well as Nikon's advanced Perfect Focus System (PFS) for thermal and vibration compensation during timelapse imaging. Both a highly-sensitive iXon EMCCD as well as a high-resolution Sona sCMOS camera are available for a wide variety of specimens and experiments.

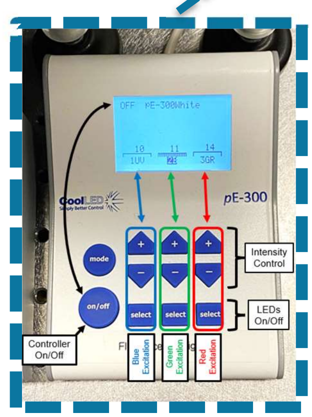
NOTE: This is a Class 4 laser-based instrument. Laser Safety training is required.

System Overview





Microscope Controls in Detail



Fluorescence Light Control in Detail

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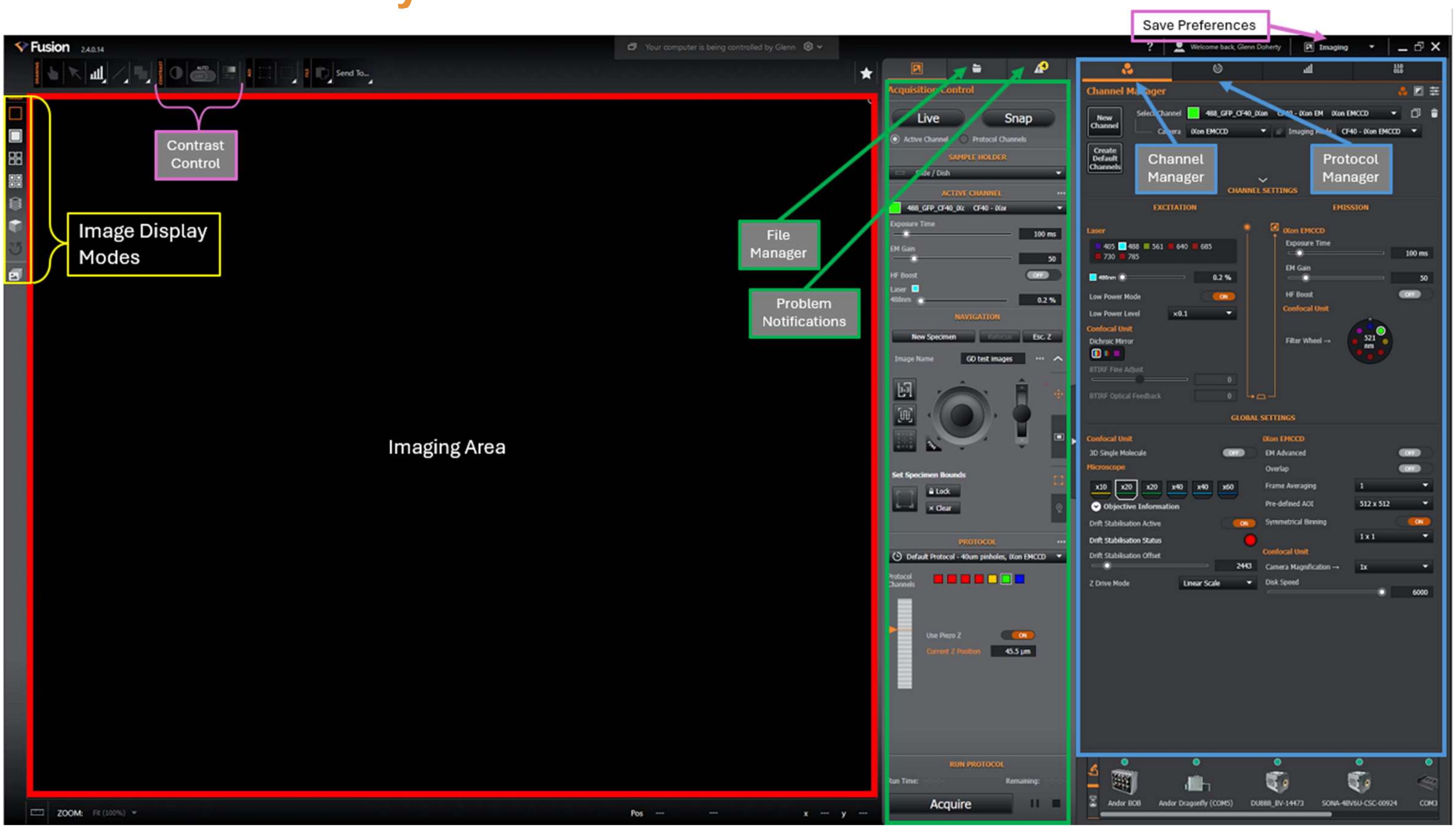
Software Layout



Imaging Area (Red) – displays the image from the Active Channel or Protocol.

Acquisition Control (Green) – simplified panel for selecting the Protocol for imaging, setting file names, relocating the sample in XYZ, making basic camera control adjustments

Expander Panel (hidden by default) – provides the user greater control over Channel-specific settings, Protocol parameters, Drift Stabilization setup, etc. Contains **Channel Manager** and **Protocol Manager**. Contains **Channel Settings** and **Global Settings**. Click the arrow at right-center of Acquisition Control to access this window.

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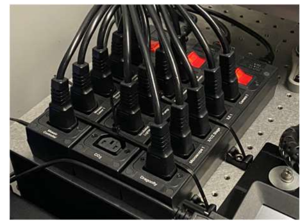
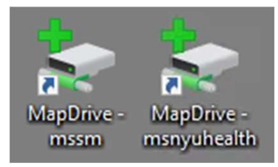
System Startup

1. Locate the three power strips. Turn on from **RIGHT to LEFT** only as needed.
    a. **Dragonfly** – fixed samples, or non-incubated samples
    b. **CO2** – live samples requiring incubation
    c. **Mosaic Controller** – for photostimulation



2. Start computer. *Log in using the credentials established at your training session.*

3. Connect to the HIVE using MapDrive.
    a. "MapDrive – mssm" for @mssm.edu emails
    b. "MapDrive – msnyuhealth" for @mountsinai.org emails



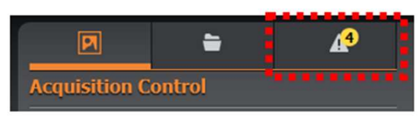
4. Before starting the software, wait for Dragonfly Status Light to change from ORANGE to GREEN.

The Status Light is located at the lower lefthand corner of the confocal unit (big, blue cover).

5. Start **Fusion**. For use of the Mosaic DMD, also start **Andor IQ3**.

Mosaic DMD setup through IQ3 is not covered in this guide. Please speak with CoRE personnel for assistance.

6. After Fusion software fully boots to the user interface, check **Problem Notifications** for any warnings.
    a. **Yellow warnings** may potentially be ignored but should be heeded (laser interlock, pixel size incorrect, etc.).
    b. **Red warnings** should be reported to CoRE personnel for investigation.



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Incubation Startup

1) Ensure the middle power strip, **CO2**, is turned on.

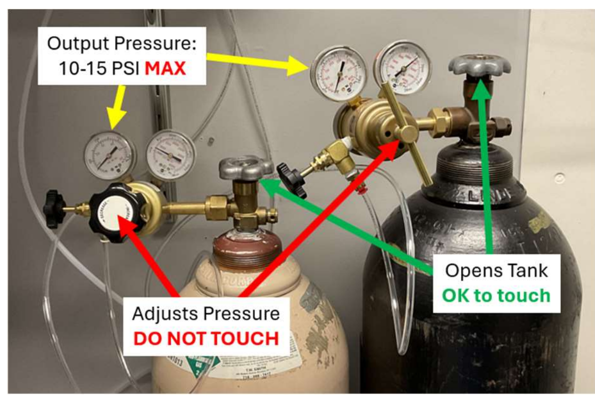
2) Press and hold the power button on the **Incubation Control** (yellow box) until the screens turns on.
3) Set the incubation parameters on the touchscreen if not already programmed:
   Temperature – 37.0C
   CO2 – 5.0%
   O2 – AIR (20%)

If other values are set or are required, tap on the appropriate parameter and make adjustments.

The chamber has reached stability when the light bar at the top of the **Incubation Control** turns green (blue box).

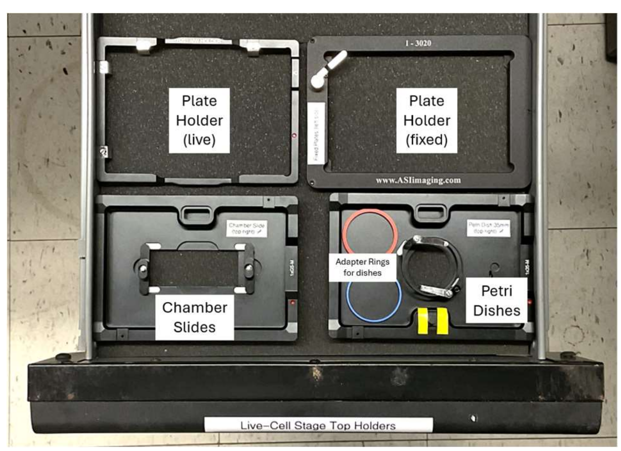
4) Double-check that the valves on the CO2 tank (biege) and Nitrogen tank (black) are open. **Open only the silver knobs over the tank – do not adjust the regulator pressure.**

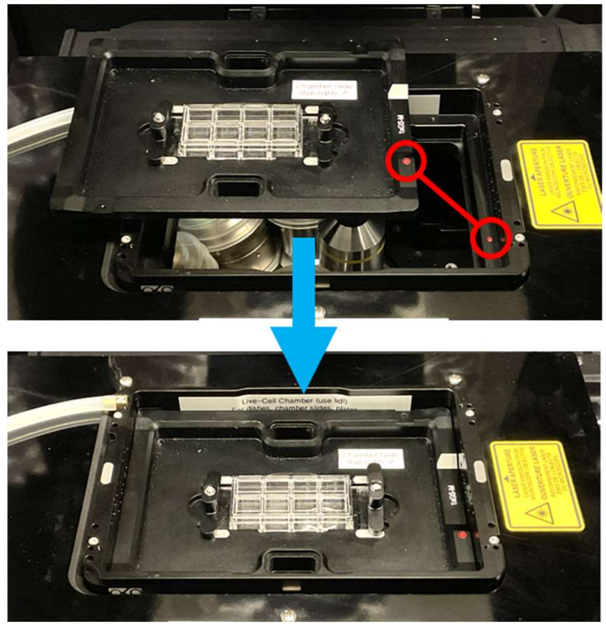
The tanks should show 10-15PSI each on the left gauge of the regulators.



5) Select the appropriate **Live-Cell Stage Holder** for your sample.

Various holders are available in the top drawer of the cabinet to the right of the air table.





6) These inserts are installed into the **Live-Cell Chamber** that is located on the left side floor of the microscope's incubation chamber.

Install the Live-Cell Chamber into the stage opening. Then, install the Stage Holder into the Chamber.

If a red dot is noted on the Holder, rotate the Holder so the dot aligns with the dot in the Chamber.

7) Keep the incubation doors of the microscope chamber closed in order to begin heating the microscope.

The Transmitted Light Arm needs to be tilted forward for the top two doors to close.

Full heating will take approximately 30 minutes to 1 hour. Anticipate this extra setup time when you book the microscope in iLabs.

For imaging during this warmup period, consider using the Drift Stabilization function to overcome thermal drift.

8) When exchanging samples, open the incubation doors only for as long as is necessary. Close them again once the sample is successfully changed.

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Prerequisites

1. **Users are responsible for cleaning and inspecting ALL objectives before AND after their session.**
    A. Open the top and front incubator doors (four in total).
    B. Tilt back the Transmitted Light Arm to gain better access to the objectives.
    C. Saturate a fresh cotton-tipped swab with Lens Cleaner.
    D. Carefully and gently wipe off the lens of the current objective. Clean the glass first and then wipe the barrel of any residue. Then, discard this swab.
    E. Take a fresh, dry swab in one hand and a clean piece of lens paper in the other. Reaching in through both front incubators doors, gently press the lens paper to the objective lens with the cotton-tipped end of the swab.
    F. Remove the paper and swab. Flip the paper over and examine for oil stains (shiny) or lens cleaner (damp/wet spot). If either are present, repeat Steps C through E using a fresh swab and more lens cleaner.
    G. Use the Objective Changer lever on the front of the microscope or the Objective listing in Fusion (under Global Settings) to switch to the next objective. Repeat Steps C through F until all objectives have been cleaned.
    H. Report any unusual scratches or substances found on the lenses to CoRE personnel immediately.
2. **All fixed samples using sealants or mountants that cure must be <u>fully dry</u> before imaging on the microscope.**

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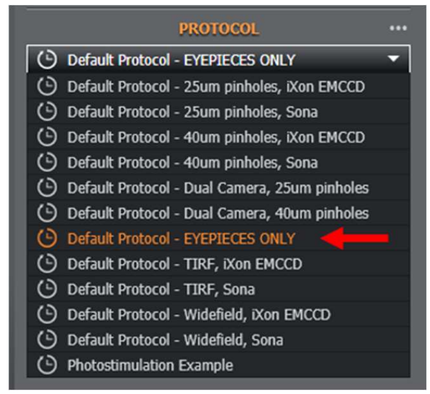
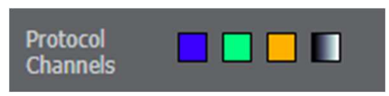
Loading a Sample and Focusing

1. In Fusion, go to **Acquisition Control > Protocol**. Select "Default Experiment – EYEPIECES ONLY."

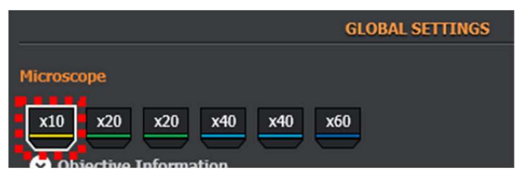


2. Select the appropriate fluorescence filter set (left to right: Blue, Green, Red) or the Brightfield setting.

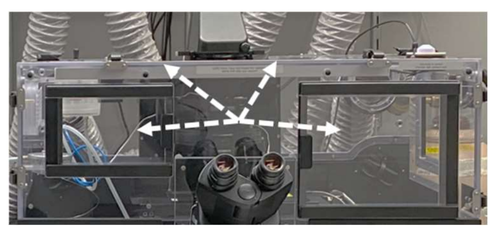


3. Expand the **Expander Panel** to access discrete microscope controls.

4. Under **Channel Manager > Global Settings > Microscope**, select the **10x** objective.



5. *On the microscope:* Slide open the doors of the incubation chamber (two in front, two on top).



6. *Microscope Controls:* Press **ESC** button. It will light up green.

7. Gently tilt back the **Transmitted Light Arm** to access the stage.

8. Check that the appropriate **stage insert** is installed. Replace as necessary. *All stage inserts are labeled based on carrier type.*

9. Load sample on to the stage insert.

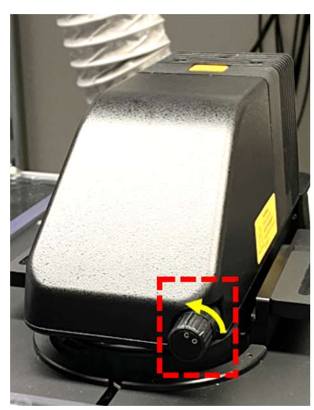
10. Gently pull **Transmitted Light Arm** forward.

11. Press **ESC** to release the objective lens for focusing. *Check that your sample is not pushed out of the stage insert!*

12. Close doors of incubation chamber. *This is recommended except when exchanging samples.*

13. Choose to activate the Fluorescence Light Control or Transmitted Light Control via the white panels.
    a. On the **Fluorescence LED control panel**, press **Select** below each of the 3 LEDs. Then press ON/OFF until the display reads ON. Adjust LED intensity individually.
    b. On the **Transmitted Light control panel**, press ON/OFF. Adjust intensity as needed (+/- buttons).

Additionally, for Transmitted Light, twist the **Transmitted Light Shutter** knob on the **Transmitted Light Arm** so that “O” is upright (“C” off to the left).



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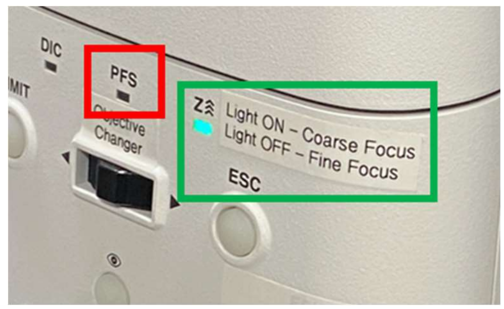
14. *On the microscope:* Press IN on the **EP1** button on right side of the microscope.

15. Look through the eyepieces and adjust focus using the **Microscope Focus Knobs**.



NOTE: Focus is **REVERSED** from most microscopes:
- Towards you >> towards the sample
- Away from you >> away from the sample

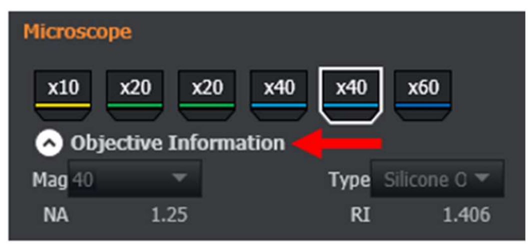
To change focus speed, tap the **Z^^^** button above the focus knob.
"Z^^^" indicator ON – coarse (fast) focus
"Z^^^" indicator OFF – fine (slow) focus

Ensure PFS Control is NOT active (light off).

16. If desired to confirm other fluorophore signals, switch to one of the other **Eyepieces** Channels via Fusion.

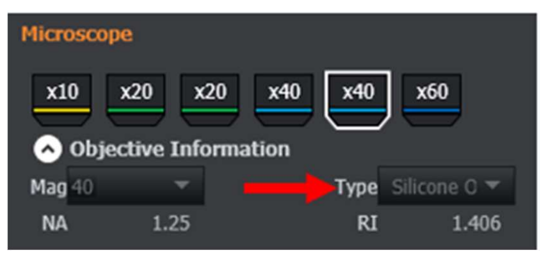
17. When focusing is complete, press **EP1**.

18. In Fusion under **Global Settings > Microscope**, switch to the desired objective lens.



Expand Objective Information to choose the appropriate lens – some lenses have the same magnification but require different immersion media.

19. If necessary, remove your sample and add a small amount of compatible immersion media on top of the lens. *<u>Confirm the appropriate media under Objective Information in Fusion.</u>*



"Silicone Oil" – use "S" immersion oil
"Oil" – use "F" immersion oil

20. Replace the sample on the stage.

21. Reconfirm correct focus using the "EYEPIECES ONLY" Protocol.

22. When switching samples, repeat all steps.

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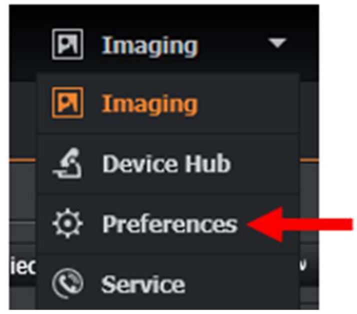
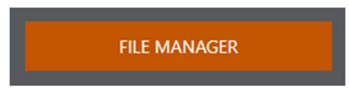
Save Preferences

These steps must be followed at the beginning of each imaging session!

1. Under the **Save** Preferences tool bar, switch from **Imaging** to **Preferences**.

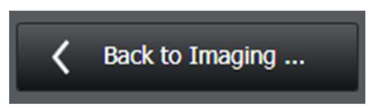


2. Click on **File Manager**.



3. Use "Browse" next to **Root Folder** to choose where the images for this session should be saved.
   a. NOTE: Remember to save to the HIVE (Y:) Drive only!
   b. NOTE: It is recommended to use None for Subfolder Creation, and also to use only the *Image Name* of the available naming options below.

4. Click **Back to Imaging**.



5. Under **Acquisition Control > Navigation**, enter an **Image Name** for the first set of images. *Images will be auto-saved and numbered sequentially.*

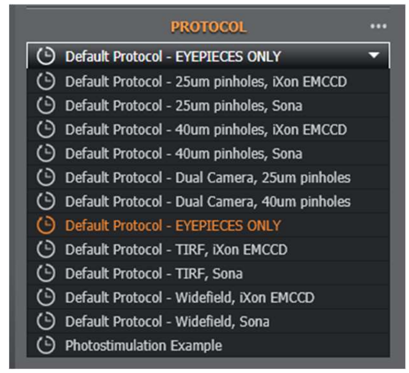
Recall Settings from Previous Images

1. Next to Acquisition Control, switch to the **File Manager** to view previously acquired images. If the image is not listed, click **Open**.
2. Double-click on the image (must be an *.ims file) that was acquired using the desired settings.
3. Under **Protocol Manager**, click **Import Settings from Image**.

PLEASE NOTE: The Fusion software does NOT recall any image properties that are listed under Channel Manager > Global Settings. You must make a separate note of these settings and manually enter them again for any repeated experiments.

Setting Up a New Protocol

1. Under **Acquisition Control > Protocol**, select the protocol for your experiment. Protocols are a combination of an Imaging Mode and Camera. Read these descriptions carefully to select the appropriate protocol.
    a. **Imaging Mode**
        i. Widefield –imaging with no pinhole/optical sectioning.
        ii. 40-micron pinholes – spinning disk mode (see description at right)
        iii. 25-micron pinholes – spinning disk mode (see description at right)
        iv. TIRF – Total Internal Reflection Fluorescence, SMLM (STORM,PALM)
    b. Camera Selection
        i. iXon EMCCD – preferable for low-signal/low-light samples. Pixel size is large, meaning resolution will suffer, but sensitivity is high.
        ii. Sona – sCMOS for high-resolution imaging. Smaller pixels means improved resolution compared to the iXon, but camera is not as sensitive to very weak signals.
        iii. Dual Camera – iXon and Sona are tied together to provide simultaneous two-color imaging. Emission signals are split at 565nm, meaning this setup is set up for red-green imaging.

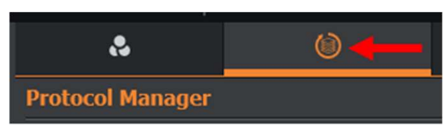


40-micron pinholes – improved sensitivity, but thicker optical section (less precise Z localization)
- *Better option for imaging with 10x, 20x, ~40x*

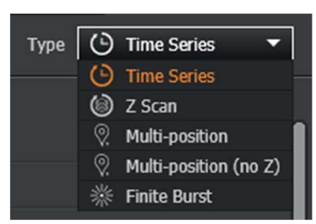
25-micron pinholes – thinner optical section (more precise Z localization), less sensitive imaging
- *Better option for imaging with 40x, 60x*

TIRF –for studying biomechanics within 100-200nm of coverglass; also used for single-molecule imaging (STORM/PALM)

2. Once you select a **Default Protocol** from the list, open the **Expander Panel > Protocol Manager**.

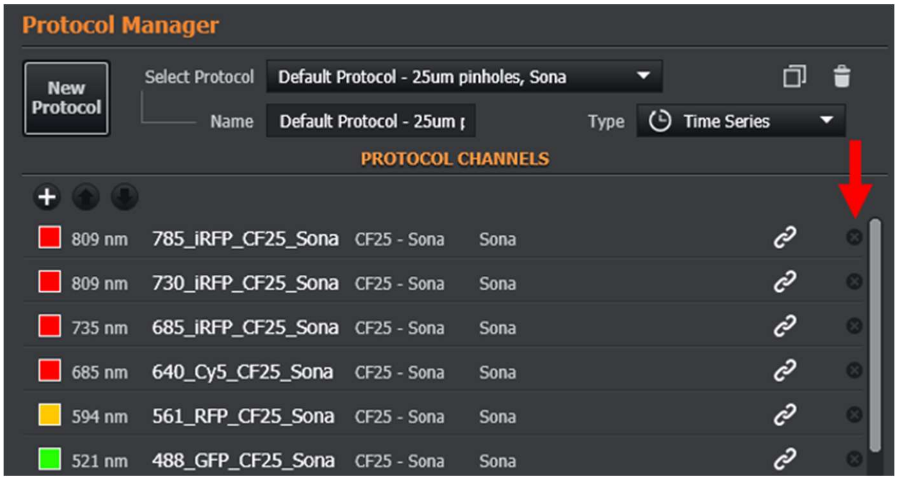


3. Select the **Type** of experiment to be performed.



In-depth descriptions about each Experiment Type are available in the next section.

4. Remove unwanted Channels from the **Channel** drop-down list. Under **Current Protocol** Channels, press the “x” button next to the channel to be removed.



NOTE: Wavelengths listed are the excitation wavelengths. Emissions are controlled by the Filter Wheel (under Channel Manager).

NOTE: 640/730 and 685/785 excitation pairs utilize different dichroic filters. For that reason, when imaging speed is a necessity, use only the following combinations of fluorophores (excitation shown):

Dichroic Position 1		Dichroic Position 2
405nm		405nm
488nm		488nm
561nm	OR	561nm
640nm		685nm
730nm		785nm

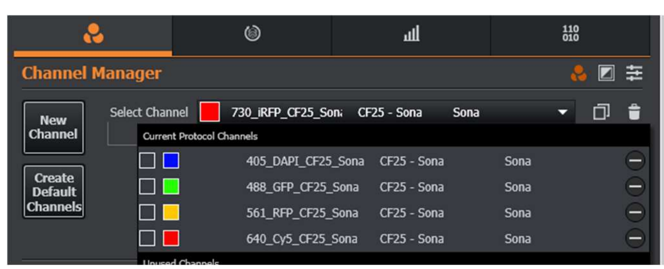
Mixing highlighted pairs of Channels (ex. 640nm and 785nm) will negatively affect imaging speeds. *Avoid having to switch dichroics if planning to study rapid cellular dynamics – choose fluorophore combinations that are ALL compatible with Dichroic Position 1 OR Dichroic Position 2.*

BY DEFAULT, 405, 488 and 561 Channels utilize Dichroic Position 1. Be certain to click through each of these Channels and switch to Dichroic Position 2 if using excitation wavelengths of 685nm and/or 785nm.

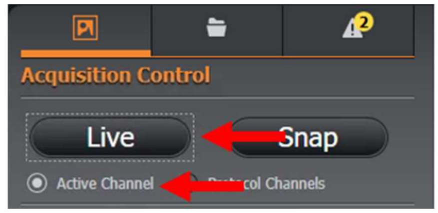
The Dichroic Position can be changed under Channel Manager > Channel Settings > Confocal Unit.

5. Switch to **Channel Manager**.

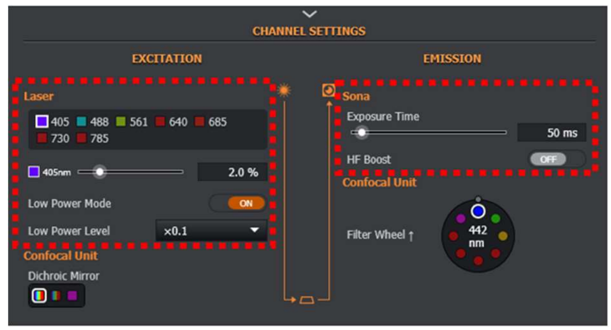
6. Use the **Select Channel** drop-down to access the first channel settings.



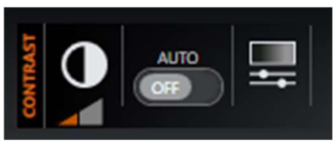
7. In **Acquisition Control**, make sure the button **Active Channel** is selected. Then, click **Live.**



8. In **Channel Settings**, adjust the Laser Power, Camera Exposure Time (and Camera Gain, if applicable) to produce a properly-exposed image.



Use the Contrast Controls found over the Imaging Area to assess proper exposure/good contrast.



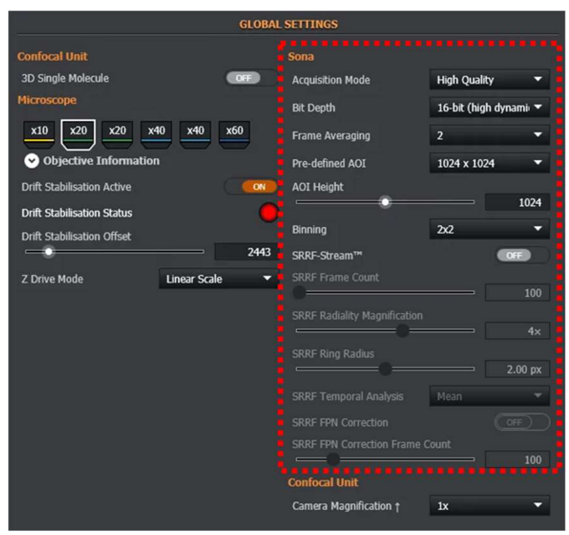
9. Change Channels from **Select Channels** and repeat Step 8.

10. As needed: In **Global Settings**, make camera adjustments to optimize image quality.

NOTE: Not all settings are available across both cameras, and not all settings are available across Protocols.

Common settings include:

1) **Pre-defined AOI** – selects a cropped Area of Interest from the camera sensor. Smaller AOIs can produce faster imaging speeds. Resolution is not affected, only recorded area size.
2) **Binning** – Groups pixels together into larger areas and displays their summed intensity across all pixels in that group. Reduces resolution but provides higher contrast and allows for faster imaging where required.
3) **Frame Averaging** – multiple image frames are captured to reduce noise. Image contrast (SNR) improves, imaging speed decreases. Good for dim or low-contrast samples.



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4) Switch to **Protocol Manager**. Set up the complete experiment parameters as required. *See next section for detailed descriptions and operation.*

5) When ready to image, click **Acquire** at the bottom of **Acquisition Control.**

Protocol Modes – In-Depth Explanations

Not all Protocol Modes are available for all experiment Types. Refer to the chart below and the in-depth descriptions to select the appropriate experiment Type. **Choose the correct column** for your experimental needs.

Protocol Mode	Experiment Type: Time Series	Z Scan	Multi-position	Multi-position (no Z)	Finite Burst
A) Time Repeat	•	•	•	•	•
B) Z Scan Settings		•	•		
C) Multi-Position Settings			•	•	
D) Montage (Fields)	•	•	•	•	
D) Montage (Edges)	•	•			
E) Drift Stabilization	•	•	•	•	•
F) Trigger Settings	•	•	•	•	•

A) Time Repeat – For time lapse imaging

a. Choose: Interval (time spacing), Frequency (images per second), or Fastest.
b. Choose: Repeat imaging for X time points or set a total imaging Duration.

B) Z Scan Settings – Capture a volume (in Z) at the current or various positions.

a. Use Piezo Z – For capturing Z-volumes that are smaller than 300 microns, activate this option for high-speed Z-stack imaging.
    i. This function must be deactivated, and objective focus used, for volumes larger than 300 microns. *This might happen with cells in suspension.*
b. Scan Mode – Start/End or Scan Center/Size.
    i. Scan Center/Size Mode is recommended for Multi-position imaging. The total Scan Size is split above and below this focal plane. The recorded Z position of each Position (under Multi-Position Settings - “Ref.Z” column) becomes the center focal plane for each Position’s scan volume.
c. Z Scan Acquisition Order –
    i. For each Z point, Acquire All Channels – the software records an image from each Channel before shifting the focal plane of the sample. *Smaller time/location displacement among Channels.*
    ii. For each Channel, acquire all Z points – the software images one complete Channel in all Z planes before repeating with the next Channel. *Faster Z-volume recording per Channel, but larger time displacement among Channels.*

(continued on next page)

d. Step Size/Step Count/Auto Step Size – Choose:
    i. The distance to refocus between Z planes (Step Size)
    ii. The total number of Z planes to capture (Step Count)
    OR
    iii. let the software decide these values based on Nyquist Sampling and either Scan Start/End or Scan Center/Scan Size (Auto Step Size)

C) **Multi-Position Settings – Capture multiple, non-contiguous fields of view.**
Can be combined with Z Scan Settings to record unique volumes at each Position.

a. Position List – shows the recorded XYZ locations that will be imaged. Use the "+" button to add a new Position after confirming the location in the Live view.
    i. When the Protocol Type "Multi-position" is used, "Ref. Z" is utilized as the Scan Center for each Position.
    ii. Drift Offset -
b. Multi-Position Protocol Order – Choose:
    i. "At each time point, visit all positions" if all independently-contained samples are running on the same time course and their kinetics should be recorded simultaneously (i.e., you wish to observe the effects across all positions at a given absolute time)
    ii. "At each position, run a time series" if independent samples should be imaged sequentially, one sample running its complete relative time course before the next sample is imaged
c. Optimize Path – for three or more Positions, the software will determine the shortest path among all Positions in order to image most efficiently.

D) **Montage – tiling/mosaic function.**
For a specific sample location, choose to image "X by Y" number of adjacent and overlapping fields. The total field area is determined by the number of tiles as well as the Field of View of the current objective lens.

a. Choose Mode:
    i. Fields – The software will image a grid of "Width" by "Height" fields. The total area by default will be centered around your current field of view, but this can be changed.
        1. "Fields" refers to the number of fields of view in X and Y to be recorded.
        2. "Relative Montage Position" tells the software if a different starting reference point for the Montage is desired.
    ii. Edges – The user defines the outermost boundaries of the imaging area exactly as desired. Use the 3x3 Montage feature (generally with a low-magnification lens like 10x/20x) in the desired Channel to acquire an overview image. Then using the Move Stage function, relocate the stage based on the sample preview to add the bounding Positions ("+" button). The total grid size will be determined by the Positions marked by the user.
b. Overlap – recommended to leave this at 10%. Increase for sparse signals, decrease for faster imaging (do not set at 0%).

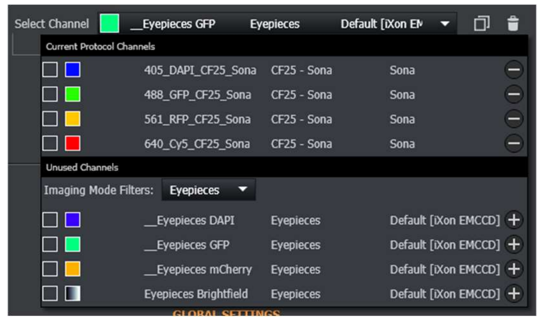
E) **Drift Stabilization Settings – Maintain stable focus over time for living specimens.**
Drift Stabilization ensures that the preferred focal plane of live specimens remains in focus for the entirety of a Time Series. It is based on the Nikon Perfect Focus System ("PFS") system which allows for the initiation of live-specimen imaging before the equipment and sample have reached thermal stability. It also corrects focus of the sample due to vibrations of the system.

The system relies on a refractive index *mismatch* between either the air and lower surface of the coverglass (for dry lenses) or between the sample and the upper surface of the coverglass. Therefore this system is best suited for samples in aqueous media (RI~1.33).

a. Apply MultiField Refocus – Allows for different imaging depths across image fields. Independent Offsets from the Reference Plane may be established that are specific to a Position. Useful when specific volumes of a cell to be imaged do not all exist at the same focal displacement from the coverglass.
    i. When used in conjunction with Multi-Position Settings and Z Scan Settings, the specific Offset for a region will act as the central focus plane of the Z-stack.
b. Enable Drift When Recording Field – When activated, focus will be corrected during image recording. This will lead to a slight decrease in recorded signal intensities. Good for very high-speed events. When deactivated, refocusing will occur only in between image recording.
c. Drift Stabilization Active – turn the PFS on or off.
d. Drift Stabilization Status – Indicates the status of the PFS.
    i. **RED** – focus not locked; PFS will not work
    ii. **ORANGE** – focus lock is being established
    iii. **GREEN** – focus has locked and is compatible with this specimen
e. Drift Stabilization Offset – When Drift Stabilization is Active, this slider must be used to establish the correct focal plane for imaging. Fusion uses the Nikon PFS to first locate the coverglass Z-plane; the user then establishes the extra Z-depth into the sample that will image the desired focal plane.

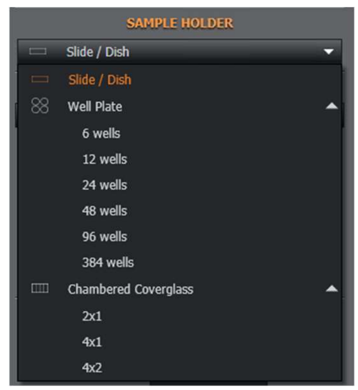
F) Trigger Settings – incorporate external event triggering into the experiment.
a. Please speak with CoRE personnel to explore options regarding this function.

Other Useful Software Features



1) For ease of switching between imaging and viewing through the eyepieces:
    a. Under **Channel Manager**, open **Select Channel**.
    b. Under **Unused Channels**, switch the **Imaging Mode Filters** to **Eyepieces**.

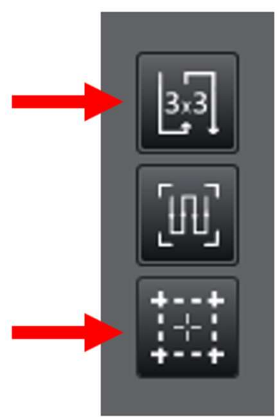
This will provide easy access to the eyepieces without having to switch Protocols.



2) For Multi-Position imaging, virtual sample carriers are available under **Acquisition Control > Sample Holder**.

After a quick calibration to align it with your real sample carrier, the virtual template can be used both to navigate among wells/chambers and also to automatically set up imaging parameters across multiple wells/chambers simultaneously. Well Navigation and Well Selection are listed under **Protocol Manager > Multi-Position Settings**.

3) 3x3 Field Preview – When locating appropriate sample fields, it is helpful to conduct a preview scan of a localized region. The "3x3" preview, available under **Acquisition Control > Navigation**, will capture a total of the nine adjacent image fields centered around the current XY stage location.



This mode can be set to capture a Montage of just the Active Channel or it can capture all Protocol Channels. Select the desired choice until **Acquisition Control (under Live)**.

This feature is best used with a low magnification objective such as 10x or 20x. Doing so will provide a large preview area of the sample. The user can then switch to a higher-magnification objective lens for final imaging.

After imaging the small Montage, use the **[--] tool (two buttons below 3x3)** and left-click on the image to relocate the stage to a desired image field, then either image the position or add it to a Multi-Position experiment.

Sample Workflows

Simple – Single position, Z-stack; fixed sample
Intermediate – Z-stack, timelapse, same focal Offset; live sample
Advanced – Multi-position, timelapse, Z-stack, Montage; live sample

These workflows assume that the correct Protocol as well as applicable Channels and Channel settings have already been chosen and set. Save Preferences should already have been established.

Simple Workflow – Single position; fixed sample

1) Set the **Experiment Type** under **Protocol Manager** to **Time Series**.
2) Locate roughly the correct position using the Eyepieces.
3) Use the **Live** image and appropriate **Channel** to produce an image in Fusion.
4) Use the **"3x3" preview** to sample the surrounding area. Use the **"[--]" function** to fine-tine the area selection (left double-click on the Montage to move the stage).
5) Click **Live** to reconfirm focus at the new location, or if the objective lens is changed.
6) Click **Acquire** to begin image capture.

Intermediate Workflow – Single position, Z-stack, time lapse; live sample

1) Set the **Experiment Type** under **Protocol Manager** to **Z Scan**.
2) Locate roughly the correct position using the Eyepieces.
3) Under **Time Repeat**, set up the parameters (frequency, duration) of the time lapse imaging course.
4) Use the **Live** image and appropriate **Channel** to produce an image in Fusion.
5) Use the **"3x3" preview** to sample the surrounding area. Use the **"[--]" function** to fine-tine the area selection (left click on the Montage to move the stage).
6) Switch to a higher-magnification objective lens if required.
7) Click **Live**.
8) Under **Z Scan Settings**, click and drag the orange Focus Plane Indicator to the desired first Z-plane for the Z-stack. Click **Start**.
9) Drag the Focal Plane Indicator so that the focus moves through the sample and to the final focal plane. Click **End**.
10) Under **Scan Mode**, switch from **Start/End** to **Centre/Size**.
11) Under **Drift Stabilization Settings**, set **Drift Stabilization Active** to **ON**.
12) Use **Drift Stabilization Offset** slider to relocate the CENTER of the specimen's volume (where you want the center of your Z-stack to be).
13) Under **Z Scan Settings**, click **Set** next to **Scan Centre**. This step aligns the focus maintenance control (Drift Stabilization) with the sample Z location.
14) Turn off **Live**.
15) Click **Acquire**.

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Advanced Workflow – Multi-position, Z-stack, Montage, time lapse, different Drift Offset; live sample

1) Set the **Experiment Type** under **Protocol Manager** to **Multi position**.
2) Locate roughly the correct position using the Eyepieces.
3) Under **Time Repeat**, set up the parameters (frequency, duration) of the time lapse imaging course.
4) Use the **Live** image and appropriate **Channel** to produce an image in Fusion.
5) Use the **"3x3" preview** to sample the surrounding area. Use the **"[--]" function** to move to the first Position (left click on the Montage to move the stage).
6) Under **Multi-Position Settings**, click **"+"** to add the Position to the Position List.
7) Click again on the preview Montage to move the stage to the next Position. Repeat Step 6.
8) Repeat Steps 6 and 7 for all remaining Positions.
9) Switch to a higher-magnification objective lens if required.
10) Under **Multi-Position Settings**, double-click on the first Position to return to it.
11) Click **Live**.
12) Under **Drift Stabilization Settings**, set **Drift Stabilization Active** to **ON**.
13) Use the **Drift Stabilization Offset** slider to relocate the center Z-plane of the Position (where you want the center of your Z-stack to be for this Position).
14) Under **Z Scan Settings > Scan Mode**, switch from **Start/End** to **Centre/Size**.
15) Click **Set** next to **Scan Centre**.
16) Set a **Scan Size** (total Z volume).
17) Under **Multi-Position Settings**, click the **Replace Position** button (to the right of +).
18) Double-click on the next Position in the Position List.
19) Repeat Steps 13, 15 and 17 to update the second Position.
20) Repeat Steps 18 and 19 for all remaining Positions.
21) Under **Montage**, click **Montage Enabled**.
22) Set the number of **Fields** (Width by Height) and set **Overlap** to **10%**.
23) Click **Acquire**.

System Shutdown

1) Ensure all images are saved to the HIVE.
2) Close the software. **"Shutting Down"** will appear on the screen. **Stay logged in and leave power strips ON until this message disappears.**
3) Clean the objective lenses.
4) Log off the computer.
5) Turn off the Incubation Control, if used, by pressing and holding the power button. Click OK to shut it down.
6) Turn off the power strips.

Instrument	**Andor Dragonfly 620**				
SN					
Current SW Version					

Objectives	**Magnification/NA**	**Immersion**	**Coverglass**	**Manufacturer PN**	**WD**
	1) PlanApo 10x/0.45 λD	air/dry	Requires "#1.5", 0.17mm coverglass	MRD70170	4.0mm
	2) PlanApo 20x/0.8 λD	air/dry	Requires "#1.5", 0.17mm coverglass	MRD70270	0.8mm
	3) Apo LWD 20x/0.95 WI λS	Water immersion	Correction collar: 0.11-0.23mm coverglass	MRD77200	0.99-0.9mm
	4) Apo LWD 40x/1.15 WI λS	Water immersion	Correction collar: 0.15-0.19mm coverglass	MRD77410	0.61-0.59mm
	5) PlanApo 40x/1.25 Sil λS	Silicone oil ("S" bottle)	2 Correction collars: 23C/37C; 0.13-0.21mm coverglass	MRD73400	0.3mm
	6) Apo TIRF 60x/1.49	Oil immersion ("F" bottle)	2 Correction collars: 23C/37C; 0.13-0.19mm coverglass	MRD01691	0.16-0.07mm

Excitation Lasers	405nm	*Excitation LED (Mosaic DMD only) >>>>*	*400nm*
	488nm		*435nm*
	561nm		*470nm*
	640nm		*500nm*
	685nm		*550nm*
	730nm		*580nm*
	785nm		*635nm*
			740nm

Detectors	Oxford Instruments iXon Life 888 EMCCD	Pixel size: 13um X 13um	Total pixels: 1024x1024	Framerate: up to 56FPS	
	Oxford Instruments Sona 4.2 back-illuminated sCMOS	Pixel size: 6.5um X 6.5um	Total pixels: 2048x2048 (2048x2046 usable)	Framerate: up to 135FPS	SRRF super-resolution imaging capable

Largest Field Size

Dichroics	405/488/561/640/730	*Dichroics allow for simultaneous*
	405/488/561/685/785	*acquisitions of multiple*
		florophores.

Techniques Available	widefield
	spinning disk confocal - 40um pinholes
	spinning disk confocal - 25um pinholes
	photostimulation; photoactivation; optogenetics
	TIRF
	FRAP (445nm laser)
	STORM (single-molecule imaging)
	Dual-camera imaging (565nm emission splitter)
Compatible with:	standard slides (76x26mm)
	Petri dishes (~35mm; adapters available)
	Well plates (6, 12 24, 48, 96, 384)
	Chambered coverglass (up to 8 wells; 4 max reccomended)
	Full incubation chamber, with variable CO2
Accessories	and heated whole microscope enclosure
	Mosaic DMD: LED- or laser-based photostimulation
	or photoablation across multiple ROIs simultaneously
	PFS - Nikon Perfect Focus System for maintaining focus
	over time lapse imaging